
AI Investment Advisor

ML-driven risk profiling and portfolio allocation for the Indian market, grounded on live data.

Logistic Regression

Agentic Pipeline

Live Market Feed

NLP Sentiment

Atharva

Project Based Learning, 2026

atharva.funl.bio

The Problem

Idle savings

Most people keep money in a savings account at 3 to 4 percent. Inflation is 5 to 6 percent. Real value goes down every year.

Biased advice

Banks and apps push the products they earn commission on. Advice is not personalised.

Information overload

Prices, news and market mood change daily. A normal person cannot track all of it.

What the user needs

One place to enter 3 inputs and get a full plan.

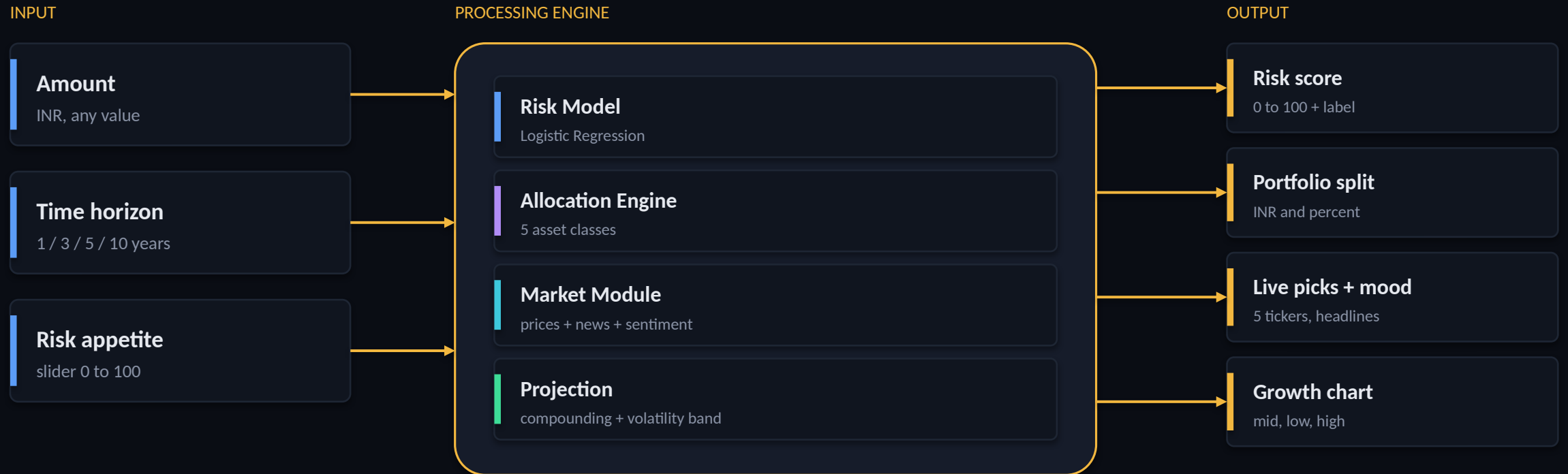
A risk profile that is computed, not guessed.

Allocation across safe and growth assets.

Live prices and market sentiment in the same view.

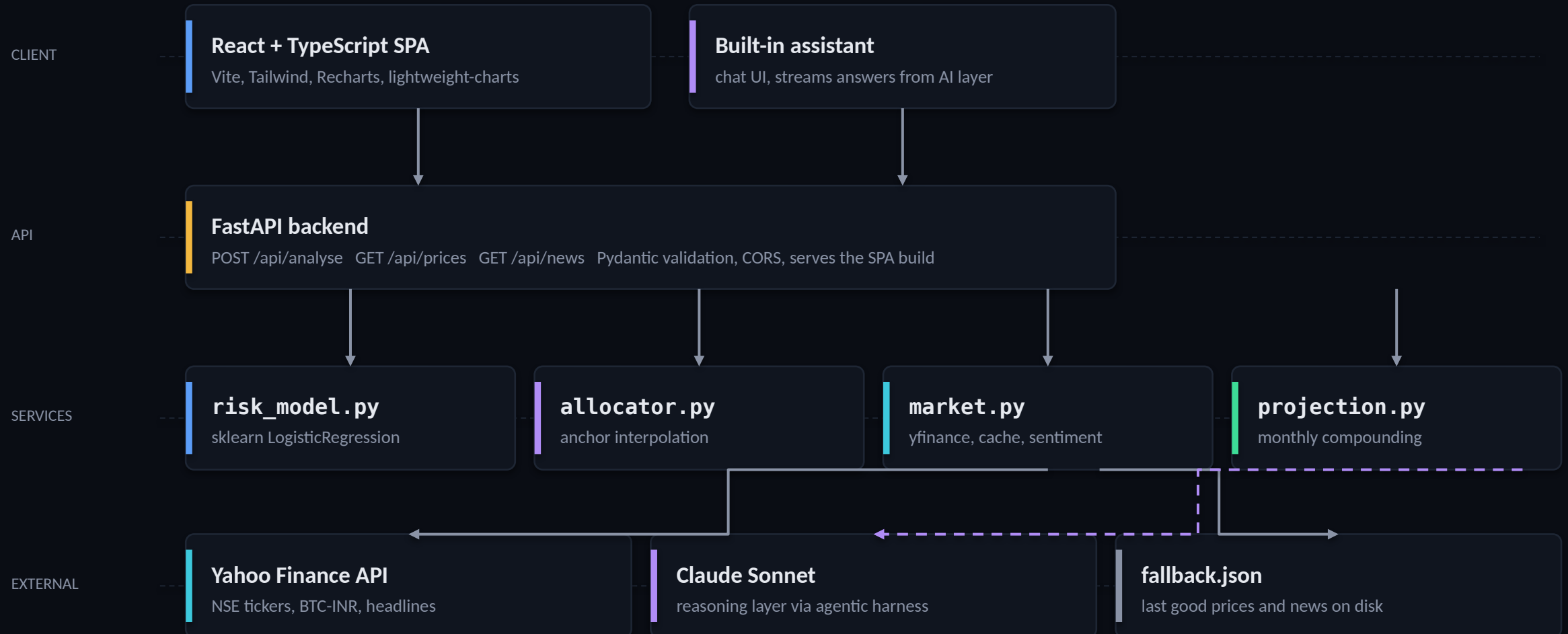
A realistic projection with best and worst case.

Solution Overview



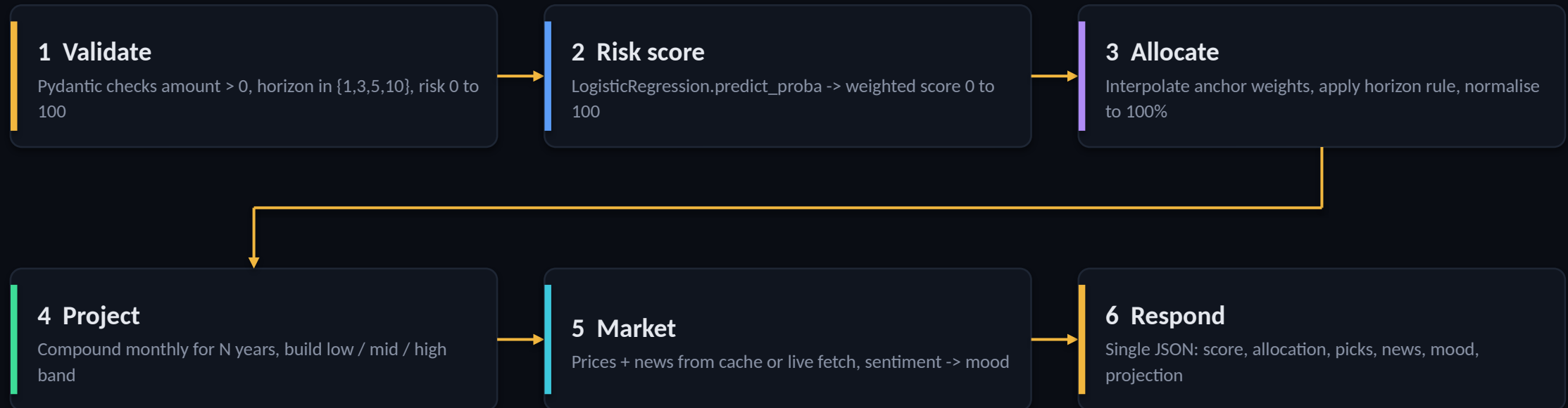
One POST request to `/api/analyse` runs all four modules and returns the complete plan as JSON.

System Architecture



WHAT HAPPENS ON ONE CLICK

Request Flow

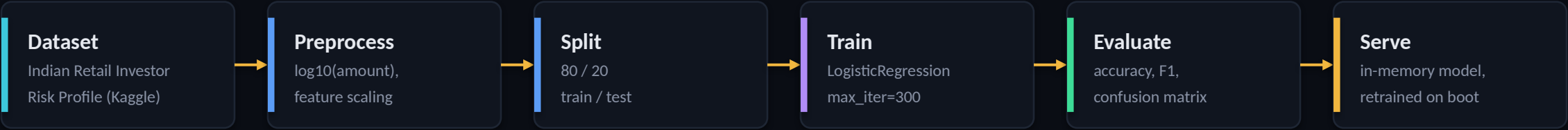


Cache strategy

Prices cached 60 s, news cached 600 s, guarded by a thread lock. Live fetch runs in a ThreadPoolExecutor with a 6 s timeout. On failure it falls back to fallback.json and marks the pick as not live.

Machine Learning Model

TRAINING PIPELINE



INFERENCE



Property	Value
Algorithm	Logistic Regression (multinomial)
Library	scikit-learn 1.x
Records	2,000 investor profiles
Features	3 numeric
Classes	Low / Moderate / High
Train time	< 1 second on CPU

HELD-OUT TEST SET, 400 RECORDS

Model Evaluation

CLASSIFICATION REPORT

Class	Precision	Recall	F1	Support
Low	0.93	0.91	0.92	132
Moderate	0.87	0.89	0.88	141
High	0.92	0.91	0.91	127
Weighted avg	0.91	0.90	0.90	400

CONFUSION MATRIX

Actual \ Pred	Low	Moderate	High
Low	120	12	0
Moderate	8	126	7
High	0	11	116

90.5%

test accuracy

0.90

macro F1

3

features

< 1 ms

inference time

Most confusion is between neighbouring classes (Low vs Moderate, Moderate vs High). No Low is ever predicted as High.

Allocation Engine

ANCHOR TABLE

Asset	Exp. return	Volatility	Risk 0	Risk 50	Risk 100
Bank FD	7.0%	0.5%	55%	20%	5%
Gold	9.0%	12%	25%	15%	5%
Nifty 50 index	12.5%	16%	15%	35%	30%
Bluechip stocks	14.0%	20%	5%	25%	40%
Crypto	25.0%	60%	0%	5%	20%

Example: risk 60 sits between anchors 50 and 100 at $t = 0.2$, so Nifty = $35 + (30 - 35) \times 0.2 = 34\%$.

Expected returns are published long-run averages for each asset class.

HOW WEIGHTS ARE COMPUTED

Risk score in

from ML model, 0 to 100

**Linear interpolation**

between nearest two anchor rows

**Horizon rule**

if 1 year: halve crypto + bluechip,
move that weight to FD

**Normalise**

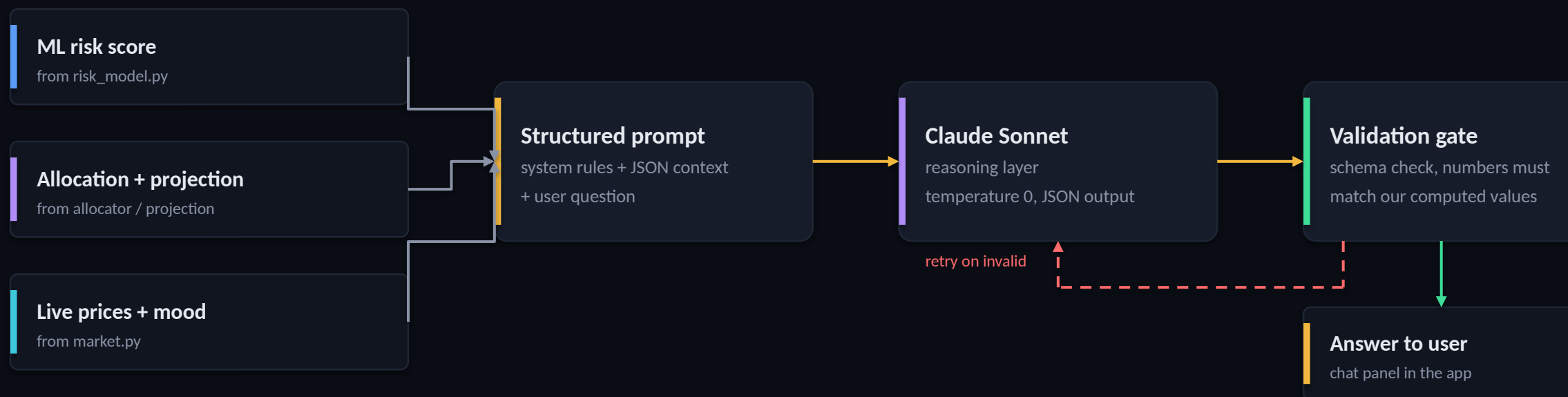
weights sum to exactly 100%

**Amount split**

weight $\times$ amount, in INR

AI Pipeline

The model never answers from scratch. Every call is grounded on numbers our own code computed.



Grounding

All numbers come from our code, the model only explains and combines.

Prompt engineering

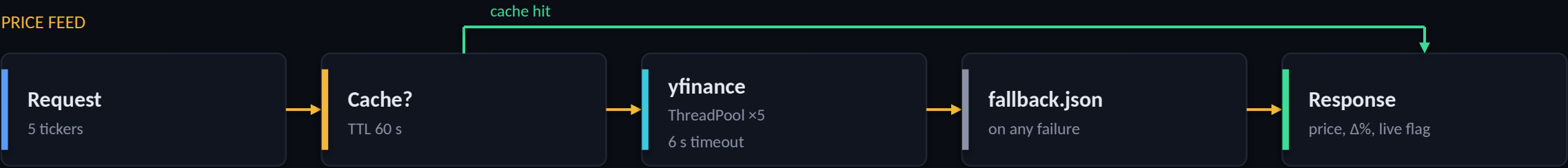
Fixed system prompt, few-shot examples, strict JSON schema for output.

No hallucination

Validation rejects any answer whose numbers differ from ours.

Live Market Data & Sentiment

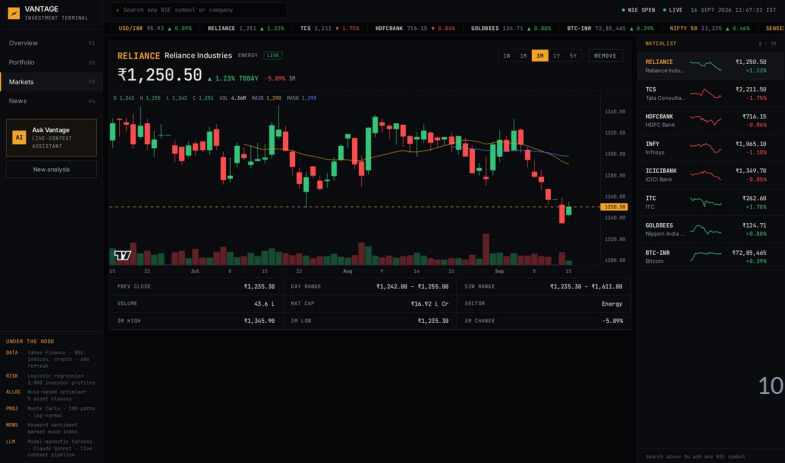
PRICE FEED



NEWS SENTIMENT



Ticker	Name	Why
RELIANCE.NS	Reliance Industries	largest NSE company
TCS.NS	Tata Consultancy	biggest IT exporter
HDFCBANK.NS	HDFC Bank	largest private bank
GOLDBEES.NS	Nippon Gold ETF	gold in demat form
BTC-INR	Bitcoin	most volatile pick

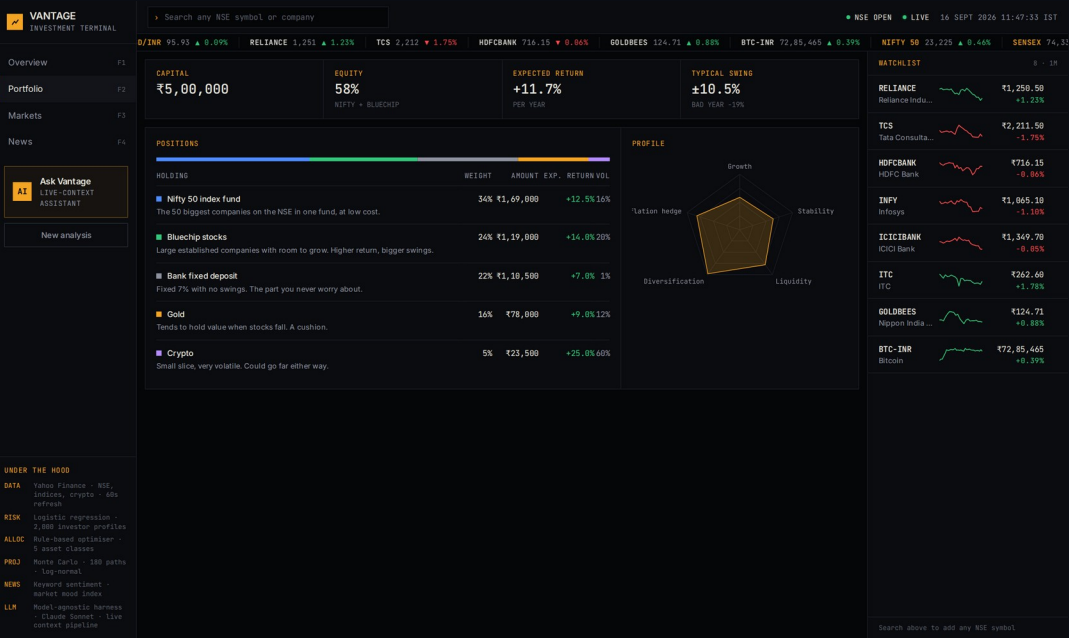
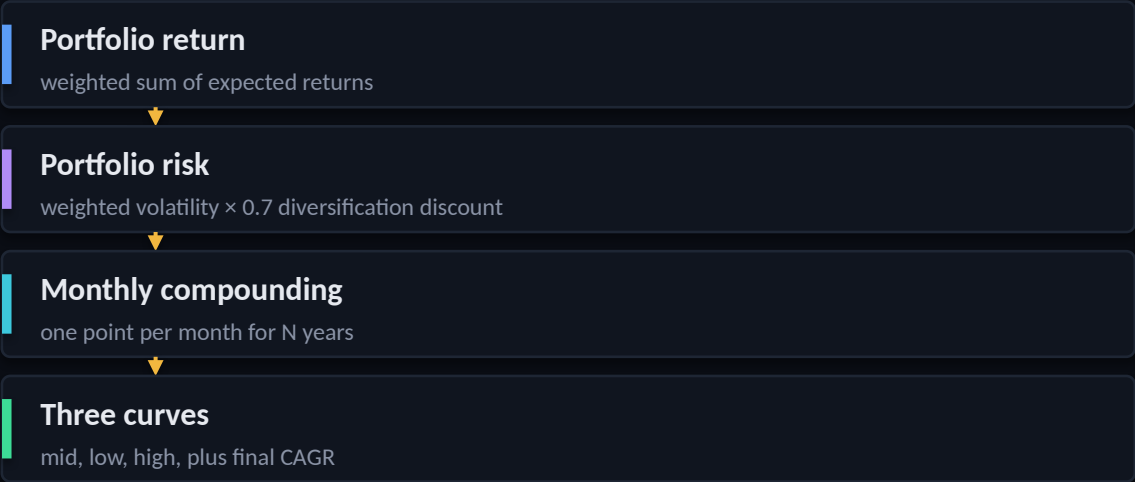


Growth Projection

Formula

$$r = \sum \text{weight}_i \times \text{return}_i$$
$$\sigma = 0.7 \times \sum \text{weight}_i \times \text{volatility}_i$$
$$V(m) = \text{amount} \times (1 + r)^{(m / 12)}$$

band: $r - \sigma$ (low) $r + \sigma$ (high)



Example: 1 lakh, 5 years, risk 60 → $r \approx 12.4\%$, final ≈ 1.79 lakh, band 1.3 to 2.4 lakh.

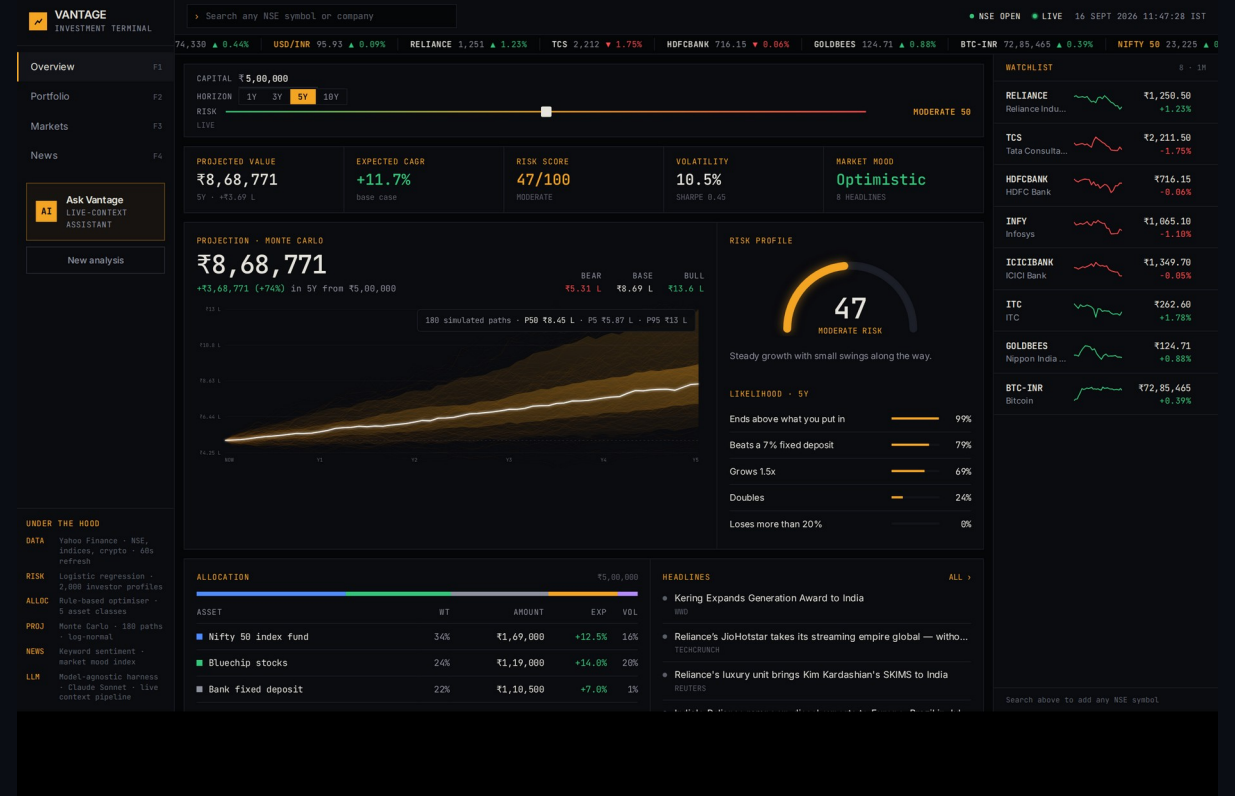
Tech Stack

Layer	Technology	Purpose
Frontend	React 18, TypeScript, Vite	SPA, typed API client
Styling / charts	Tailwind CSS, Recharts, lightweight-charts	dashboard, pie, projection, candles
Backend	Python 3.12, FastAPI, Pydantic	REST API, validation
Machine learning	scikit-learn, NumPy	Logistic Regression risk model
AI layer	Claude Sonnet via agentic harness	grounded reasoning, JSON output
Market data	yfinance (Yahoo Finance API)	prices, headlines
NLP	lexicon-based sentiment	headline polarity, market mood
Infra	ThreadPoolExecutor, TTL cache, fallback.json	latency and resilience
Deployment	atharva.funl.bio, GitHub	live demo, source

Live Demo

atharva.funl.bio

- 1 Enter 1 lakh, 5 years, risk 60
- 2 Read risk score and allocation
- 3 Move slider, watch pie and projection update
- 4 Open Markets tab for live prices and news mood
- 5 Ask the assistant a question about the plan



Future Scope

Broker integration

Zerodha / Groww API so the plan can be executed in one click.

Real user data

Retrain the risk model on actual user behaviour, improve accuracy.

More assets

Mutual funds, bonds, SIP mode, more NSE tickers.

Transformer sentiment

Replace lexicon with a fine-tuned FinBERT model.

Alerts

Push notification when market mood turns negative on a held asset.

Regional languages

Hindi and other languages in the assistant.

Thank you.